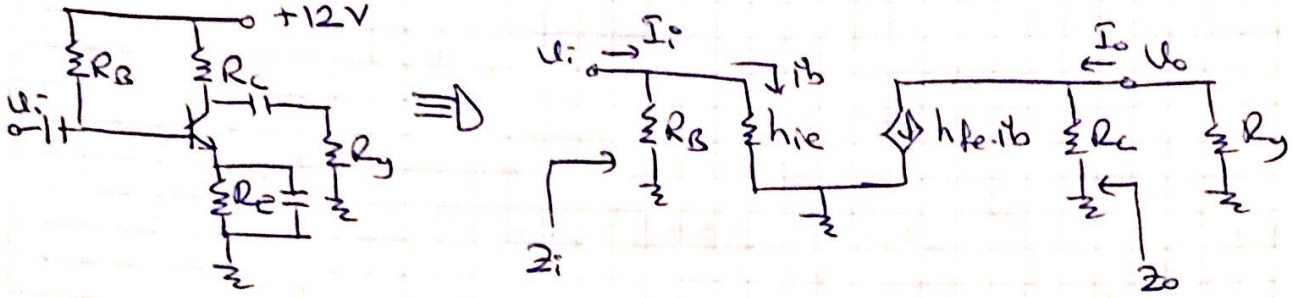


BÖLÜM-2

TEK KATLI TEMEL YÜKSELTEÇ DEVRELERİNİN ANALİZİ (BJT)

1. SABİT KUTUPLANALI YÜKSELTEÇ:



$$Z_i = R_B \parallel h_{ie}$$

$$Z_o = R_C$$

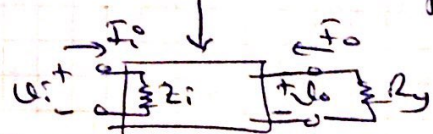
$$U_o = -h_{fe} \cdot i_b (R_C \parallel R_L) \Rightarrow A_v = -\frac{h_{fe}}{h_{ie}} (R_C \parallel R_L)$$

$$I_o = h_{fe} \cdot i_b + \frac{U_o}{R_C} = h_{fe} \cdot i_b - \frac{h_{fe} \cdot i_b (R_C \parallel R_L)}{R_C} = \frac{h_{fe} R_C R_L}{R_C + R_L} i_b$$

$$I_i = i_b + \frac{U_o}{R_B} = i_b + \frac{i_b h_{ie}}{R_B} = i_b \left(\frac{R_B + h_{ie}}{R_B} \right)$$

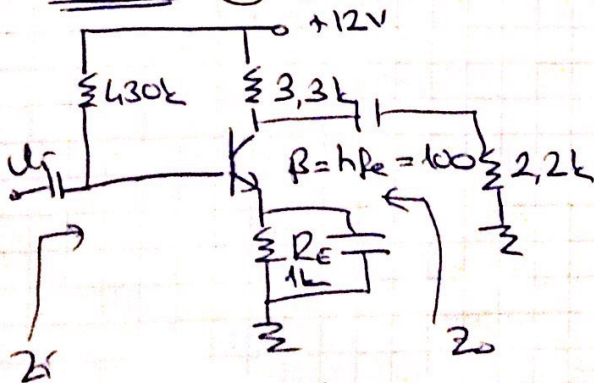
$$A_i = \frac{I_o}{I_i} = \frac{h_{fe} R_C R_L}{(R_C + R_L)(R_B + h_{ie})}$$

$$2. \text{ci} \text{ y} \text{ol} = A_i = -A_v \cdot \frac{R_B}{R_L}$$



$$A_i = -\left(\frac{h_{fe}}{h_{ie}} \cdot \frac{R_C \cdot R_L}{R_C + R_L} \right) \cdot \frac{R_B \parallel h_{ie}}{R_L} = \frac{h_{fe} R_C R_L}{(R_C + R_L)(R_B + h_{ie})}$$

Örnek: *



a) $r_e = ?$

b) h_{ie} modeli ile Z_i , Z_o , A_v ve A_i 'yi bulunuz.

Gözetim:

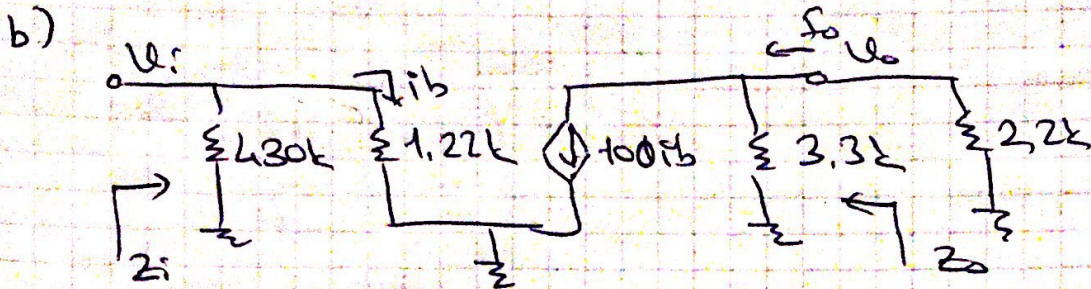
$$a) \quad 12 = 430k \cdot I_B + 0,7 + (101) \cdot 1k \cdot I_B = 0$$

$$\Rightarrow I_B = \frac{11,3}{430k + 101k} = 21,28 \mu A \Rightarrow I_E = 101 \cdot 21,28 \mu A$$

$$I_E = 2,15 mA$$

$$r_e = \frac{26mV}{2,15mA} \approx 12,1 \Omega$$

$$\beta r_e = 1,22k\Omega$$



$$Z_i = 430k \parallel 1,22k$$

$$Z_i = 1,21k\Omega \quad Z_o = 3,3k\Omega$$

$$A_{v_e} = - \frac{h_{fe}}{h_{ie}} \cdot (R_c \parallel R_L) = - \frac{100}{1,22k} \left(\frac{3,3k \cdot 2,2k}{5,5k} \right)$$

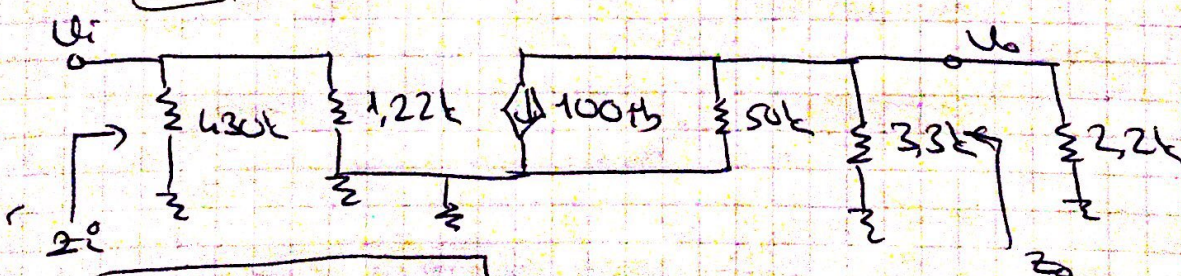
$$A_{v_e} \approx -108,28$$

$$A_v = -A_{v_e} \cdot \frac{Z_i}{R_L} = 108,28 \cdot \frac{1,21k}{2,2k}$$

$$A_v = 59,50$$

$$A_v = \frac{h_{fe} \cdot R_c \cdot R_L}{(R_c + R_L)(R_L + h_{ie})} = \frac{100 \cdot 3,3k \cdot 430k}{5,5k \cdot 431,22k} = 59,83$$

h_{oe} verken: ($\frac{1}{20ms} = 50k \cdot 1k$ bir ro olar.). $h_{oe} = 20\mu S$ verilm.)



$$Z_i = 1,21k \text{ (ayn.)}$$

$$Z_o = 3,3k \parallel 50k = 3,09k\Omega$$

$$U_o = -100 \mu b. \left(\frac{50k \parallel 3,3k}{3,09k} \parallel 2,2k \right) = -100 \mu b. \cdot 1,28k$$

$$U_i = 1,22k \cdot \mu b$$

$$\Rightarrow A_{v_e} = - \frac{100 \mu b \cdot 1,28k}{1,22k \cdot \mu b} \Rightarrow A_{v_e} = -105,33$$

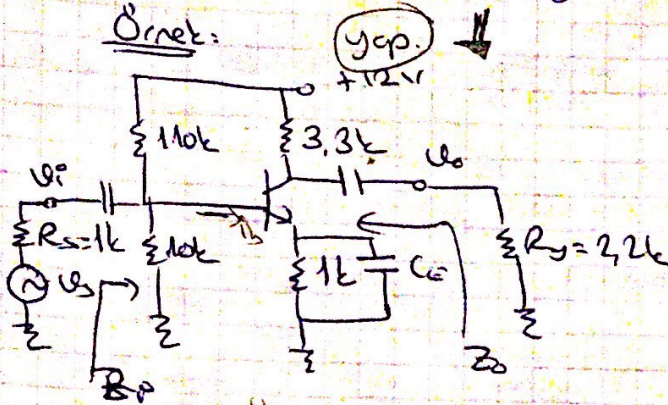
$$A_v = -A_{v_e} \cdot \frac{Z_i}{R_L} = 105,33 \cdot \frac{1,21k}{2,2k} = 57,93$$

↗ (2)

2. GERİLİM BÖLÜCÜLÜ YÜKSELTEÇ:

Sabit kutuplanmalı devre ile aynı yöntemle çözülür. Sadece R_{B2} yerine $R_{B2} = R_1 || R_2$ yazılır.

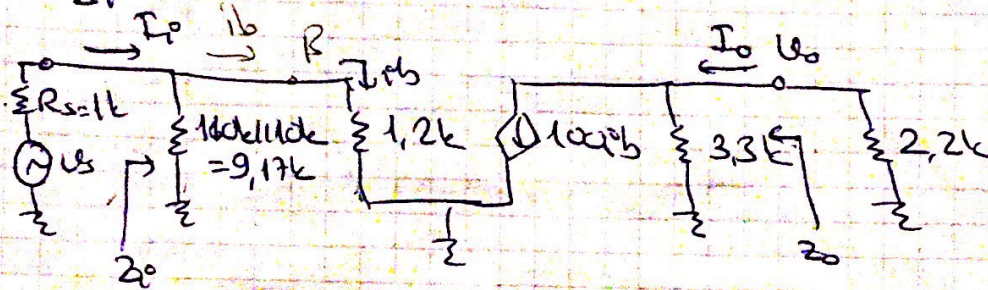
Örnek:



$$h_{ie} = 1,2k$$

$$h_{fe} = 100$$

$$Z_i, Z_o, A_v, A_{v_s} \text{ ve } A_{v_s} = \frac{V_o}{V_s} = ?$$



$$Z_o = 9,17k || 1,2k = 1,06k \quad Z_o = 3,3k$$

$$A_v = - \frac{h_{fe}}{h_{ie}} \cdot R_c || R_L = - \frac{100}{1,2k} (3,3k || 2,2k) = -110$$

$$A_o = \frac{I_o}{I_i}$$

$$I_o = I_{CQ} + \frac{V_o}{3,3k} = 100\mu A + \frac{V_o}{3,3k} = 100\mu A - 40\mu A = 60\mu A = 53,09$$

$$I_i = I_B + \frac{V_i}{9,17k} = I_B + \frac{1,2k I_B}{9,17k} = 1,13 I_B$$

$$A_{v_s} = \frac{V_o}{V_s} \Rightarrow \frac{V_o}{V_i} \cdot \frac{V_i}{V_s} \Rightarrow A_{v_s} = A_v \cdot \frac{Z_i}{Z_i + R_s}$$

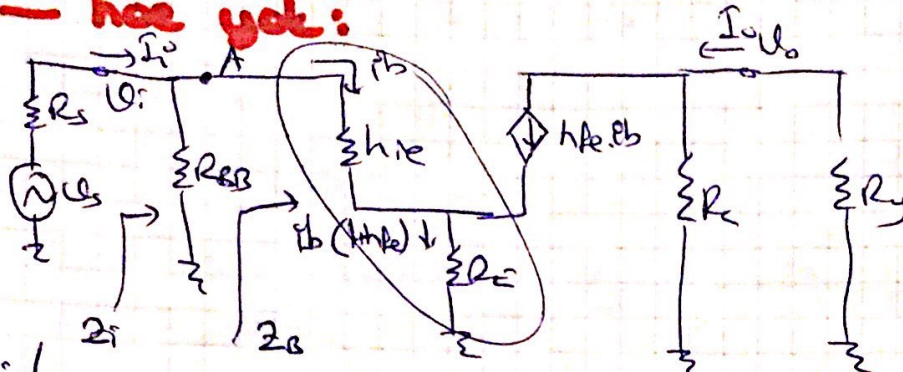
$$A_{v_s} = -110 \cdot \frac{1,06k}{1,06k + 1k} = -56,60$$

Kaynak devresinin etkisi



3. EMİTÖR KÜPRÜLENMENİP YÜKSELTEÇ (CE YOL)

— hce yol:



① Z_i Z_B dır: A ile toprak arasında bakıldığında görülen dırır.

$$Z_B = \frac{V_A}{I_B} = \frac{U_i}{I_B} = \frac{I_B \cdot h_{ie} + I_B (1+h_{fe}) \cdot R_E}{I_B} = \boxed{h_{ie} + (1+h_{fe}) R_E}$$

$Z_i = R_{EB} \parallel Z_B$ ile bulunur.

② A_v $\left. \begin{array}{l} U_i = I_B \cdot Z_B \\ U_o = -h_{fe} \cdot I_B \cdot (R_C \parallel R_L) \end{array} \right\} \boxed{A_v = -h_{fe} \cdot \frac{(R_C \parallel R_L)}{Z_B}}$

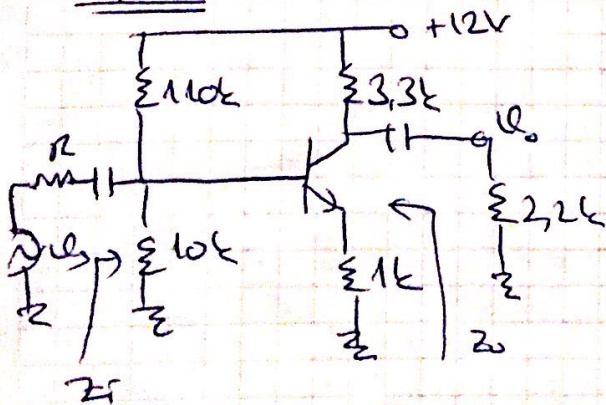
③ A_i $A_i = \frac{I_o}{I_i} = \frac{-U_o / R_L}{U_i / Z_i} = \boxed{-A_v \cdot (Z_i / R_L)}$

④ Z_o görülmeye yazılırsa; (U_s kuadere)

$(R_s \parallel R_{EB}) I_B + h_{ie} I_B + (1+h_{fe}) I_B \cdot R_E = 0$ bu eşitlik ($I_B=0$) olursa sağlanır. Bu nedenle Z_o bulunurken $I_B=0$ olur ve bağımsız akım kaynağı açık devre olur.

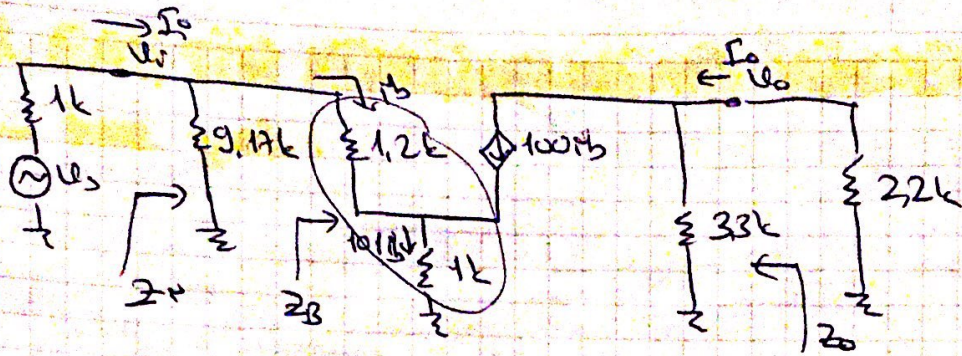
$V_E \Rightarrow \boxed{Z_o = R_E}$

Örnek:



$h_{ie} = 1,2k$
 $h_{fe} = 100$

$Z_i, Z_o, A_v, A_{v_s}, A_i = ?$



$$Z_B = 1,2k + 101 \cdot 1k \quad \left(\frac{V_i}{I_B} = \frac{1,2k \cdot I_B + 101 I_B \cdot 1k}{I_B} \right)$$

$$Z_B = 102,2k$$

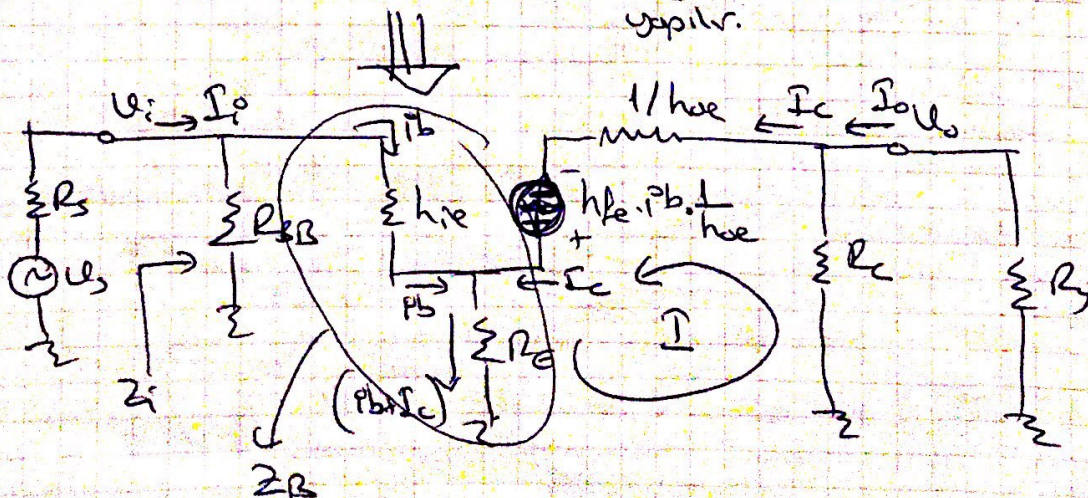
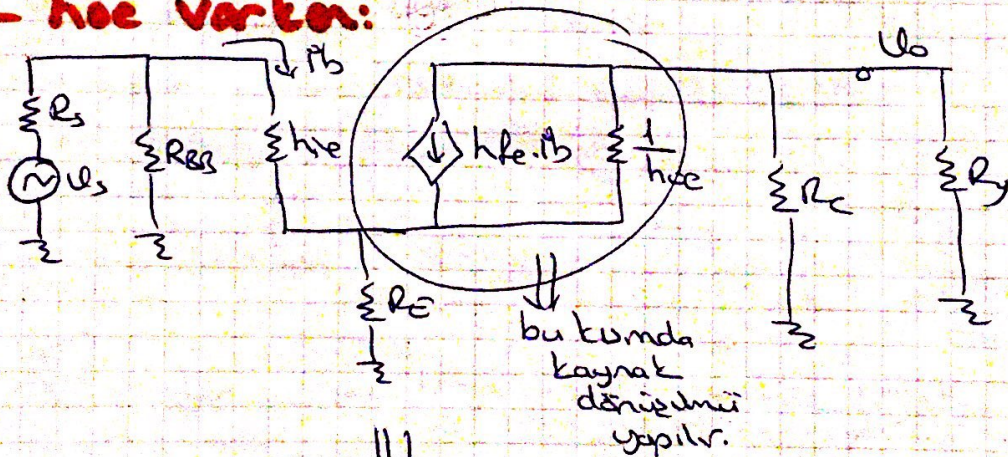
$$Z_P = 9,17k \parallel 102,2k = 8,41k \quad Z_i = 8,41k \quad Z_o = 3,3k$$

$$A_v = - \frac{100\mu A \cdot (3,3k \parallel 2,2k)}{I_B \cdot Z_B} = - \frac{132\mu A}{102,2k} = -1,29 = A_{ve}$$

$$A_{vs} = -1,29 \cdot \frac{Z_i}{Z_i + R_s} = -1,29 \cdot \frac{8,41k}{8,41k + 1k} = -1,15 = A_{vs}$$

$$A_p = -A_{ve} \cdot \frac{Z_i}{R_s} = 1,29 \cdot \frac{8,41k}{2,2k} = 4,93 \rightarrow A_p = 4,93$$

— hoe varken:



① Z_i

$$Z_i = \frac{U_i}{I_b} = \frac{h_{ie} \cdot I_b + R_E (I_b + I_c)}{I_b}$$

Gere denk.:

$$R_E (I_b + I_c) + (R_C || R_L) I_c + \frac{1}{h_{oe}} \cdot I_c - \frac{h_{fe} \cdot I_b}{h_{oe}} = 0$$

→ Bu denklemi çözümü için

bu I_c 'nin de I_b cinsinden bulunması lazım. Bunun için devrenin çıkış tarafında gerre yazılır (I nolı gerre)

$$I_c = \frac{h_{fe} - R_E h_{oe}}{1 + h_{oe} (R_E + R_C || R_L)} I_b$$

çıktır.

$$Z_i = \frac{h_{ie} + I_b + R_E (I_b + A I_b)}{I_b} = \boxed{h_{ie} + R_E (1 + A)} \quad A \text{ desek.}$$

bulunur.

$Z_i = R_{S1} || Z_i$ ile de Z_i bulunur.

② A_v

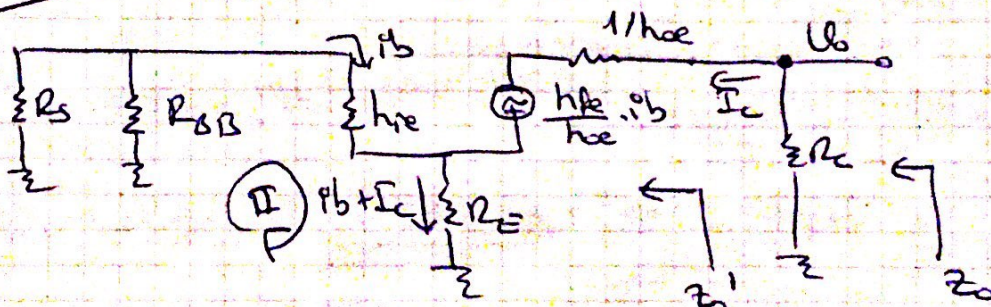
$$U_o = -I_c \cdot (R_C || R_L) = -A \cdot I_b \cdot (R_C || R_L)$$

$$U_i = I_b \cdot Z_i$$

$$\Rightarrow A_v = \boxed{-\frac{A (R_C || R_L)}{Z_i}}$$

③ Z_o

Z_o bulmak için esdeğer devre:



$$Z_o' = \frac{U_o}{I_c} = \frac{I_c \cdot 1/h_{oe} - \frac{h_{fe}}{h_{oe}} \cdot (I_b) + R_E (I_b + I_c)}{I_c}$$

Gere denk. den:

$$[(R_S || R_{B1}) + h_{ie}] \cdot I_b + I_b R_E + I_c R_E = 0$$

burada: I_b ile I_c arasında bulunsa denklem çözülür. Bunun için girer tarafında gerre yazılır. (II nolı gerre)

$$I_b = -\frac{R_E}{R_E + [h_{ie} + (R_S || R_{B1})]} I_c$$

B desek.

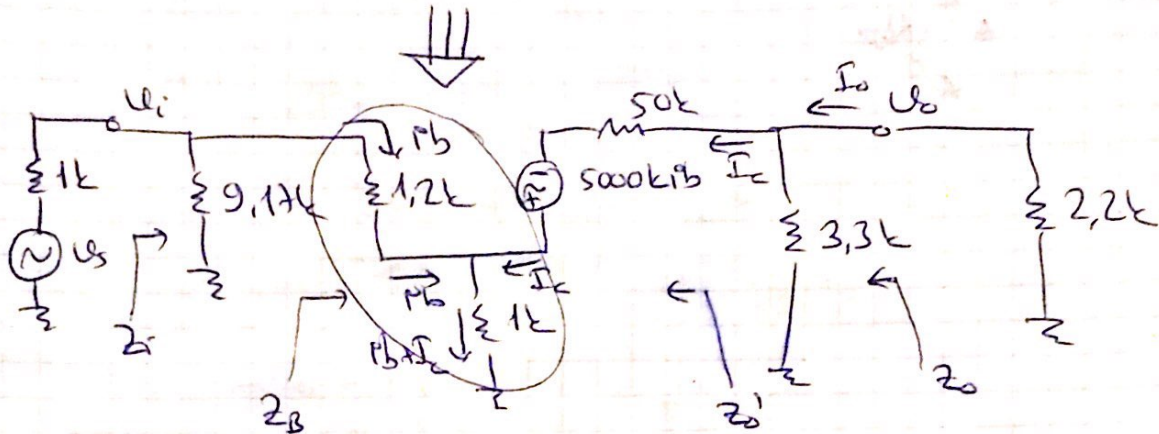
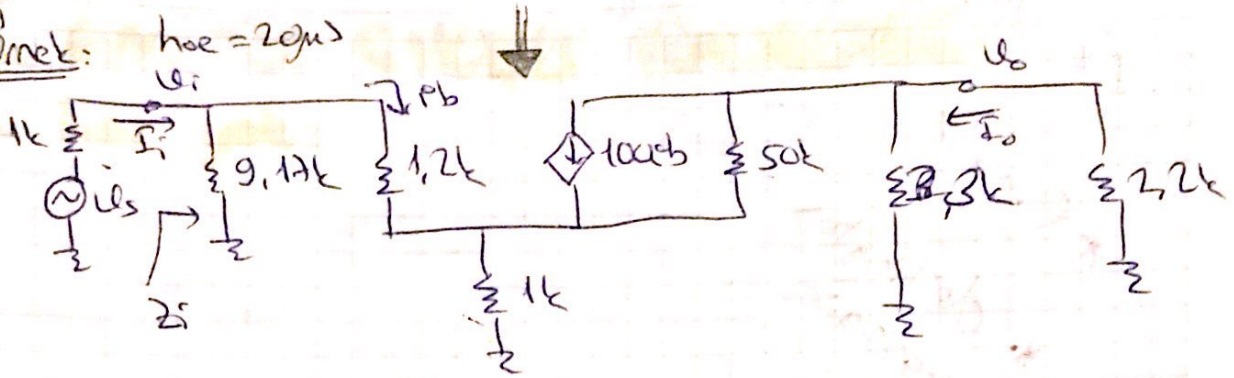
$$Z_o' = \frac{I_c \cdot 1/h_{oe} - \frac{h_{fe}}{h_{oe}} \cdot B I_c + R_E \cdot I_c (1 + B)}{I_c}$$

$$\boxed{Z_o' = \frac{1}{h_{oe}} - \frac{h_{fe} \cdot B}{h_{oe}} + R_E (1 + B)}$$

bulunur $Z_o = R_C || R_L$ bulunur.

⑥

Örnek: $h_{oe} = 20 \mu S$



$$Z_B = \frac{1,2k \parallel 1k (I_B + I_C)}{I_B} \rightarrow \text{sağ tarafta çevre: } 1,32k$$

$$1k (I_B + I_C) + (3,3k \parallel 2,2k) \cdot I_C$$

$$+ 50k I_C - 5000k I_B = 0$$

$$I_B (1k - 5000k) + I_C (1k + 1,32k + 50k) = 0$$

$$I_C = \frac{4999k}{52,32k} I_B \Rightarrow I_C = 95,54 I_B$$

$$Z_B = \frac{1,2k \parallel 1k + 1k \cdot 95,54 I_B}{I_B} = 97,74k\Omega$$

$$Z_i = 9,1k \parallel 97,74k = 8,38k\Omega \Rightarrow Z_i$$

$$U_o = -I_C \cdot (3,3k \parallel 2,2k) = -95,54 I_B \cdot 1,32k$$

$$U_i = I_B \cdot Z_i = 97,74k \cdot I_B$$

$$A_v = \frac{U_o}{U_i} = \frac{-95,54 \cdot 1,32k}{97,74k}$$

$$A_v = -1,29$$

$$A_{v_s} = A_v \cdot \frac{Z_i}{Z_i + R_s} = -1,29 \cdot \frac{8,38k}{9,38k}$$

$$A_{v_s} = -1,15$$

$$A_p = -A_v \cdot \frac{Z_p}{R_y} \Rightarrow A_p = 1,29 \cdot \frac{8,38k}{2,2k} = 4,91$$

7